RESEARCH PORTFOLIO / OCTOBER 2026

OpenMC Parametric Neutronics Simulation Study

Technical report | OpenMC 0.16.0, Python, NumPy, Pandas, Matplotlib

28	4	1,000,000
main sweep cases	boron control cases	active histories per main case

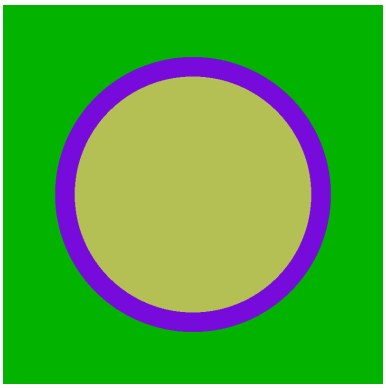
Abstract

A Python-driven OpenMC workflow studied how uranium-235 enrichment and moderator density influence neutron multiplication in a simplified, reflective pressurized-water-reactor (PWR) fuel pin cell. Across seven enrichment values (2.0–5.0 wt%) and four moderator densities (0.65–0.80 g/cm³), calculated k-effective ranged from 1.096075 to 1.353420; the largest reported standard Monte Carlo uncertainty was 0.001151. The density response varied with enrichment. Four boron-free control cases supported an important contribution from boron absorption to that response.

Model and purpose

The model was adapted from OpenMC's bundled PWR pin-cell example [1]. It retains physically meaningful fuel, cladding, and moderator regions while remaining tractable on a personal computer. The software objective was to demonstrate reproducible model generation, Monte Carlo execution, statistical assessment, CSV extraction, and scientific visualization.

Property	Study model
Fuel	UO_2; 10.29769 g/cm³; variable U-235 enrichment
Cladding	Zircaloy; 6.55 g/cm³
Moderator	Borated water; 0.65–0.80 g/cm³
Geometry	Fuel radius 0.39218 cm; cladding outer radius 0.45720 cm; pitch 1.26 cm
Boundaries	Reflective lateral boundaries; axially infinite



Pin-cell cross section. The circular fuel is enclosed by cladding and a square moderator region. Material colors do not represent neutron flux, temperature, or reaction rate. No explicit fuel–cladding gap or finite reactor core is modeled.

Scientific scope: educational sensitivity study only; not a full-core calculation, reactor digital twin, design validation, or safety analysis.

METHODS AND NUMERICAL CHECKS

Simulation design

A configurable Python function generated OpenMC materials, geometry, k-eigenvalue settings, and cell-filtered tallies. The primary study used 150 batches, 50 inactive batches, and 10,000 particles per batch, or 1,000,000 active particle histories per case. Distinct recorded seeds and directories separated the 28 main cases. Nuclear data were supplied by a local ENDF/B-VIII.1 HDF5 library.

Scores: combined k-effective, fuel flux, moderator flux, and fuel fission reaction rate. Flux tallies are source-normalized track lengths integrated across selected cells; fission scores are reactions per source particle [2]. Neither is an absolute physical rate without further normalization.

Input and implementation verification

Input assertions checked the intended 2.40% and 3.50% enrichments and moderator densities of 0.740582 and 0.650000 g/cm³. An explicitly constructed fixed-isotope model reproduced the bundled example at 1.15881 ± 0.00265 under the same inputs and random sequence. This confirms implementation consistency, not external physical validation. The later parameterized fuel uses OpenMC's enrichment correlation, so its isotope composition is not identical to the fixed-isotope reference.

Numerical precision checks

Active histories	k-effective	Statistical σ
90,000	1.161306	0.003204
400,000	1.160142	0.001423
1,000,000	1.159147	0.000977

Three independent 1-million-history replicates (seeds 101, 202, 303) returned 1.161331 ± 0.000897, 1.159469 ± 0.000937, and 1.160110 ± 0.000812. Their arithmetic mean was approximately 1.160303. Independent estimates were statistically compatible.

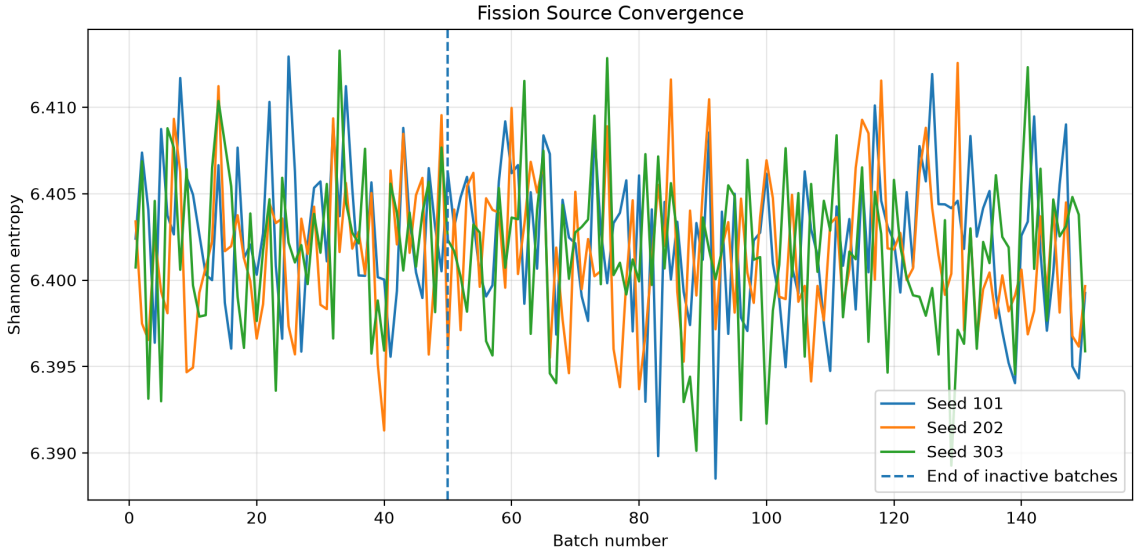


Figure 1. Fission-source Shannon entropy for three independently seeded runs; dashed line marks the end of 50 inactive batches. The curves show no persistent drift, supporting baseline source convergence [3].

PRIMARY PARAMETER STUDY

The 7 × 4 sweep covered enrichments of 2.0–5.0 wt% in 0.5%-point increments and moderator densities of 0.65, 0.70, 0.75, and 0.80 g/cm³. Every computed series showed an increase in k-effective with fuel enrichment. Error bars in the plots represent the reported one-standard-deviation Monte Carlo sampling uncertainty.

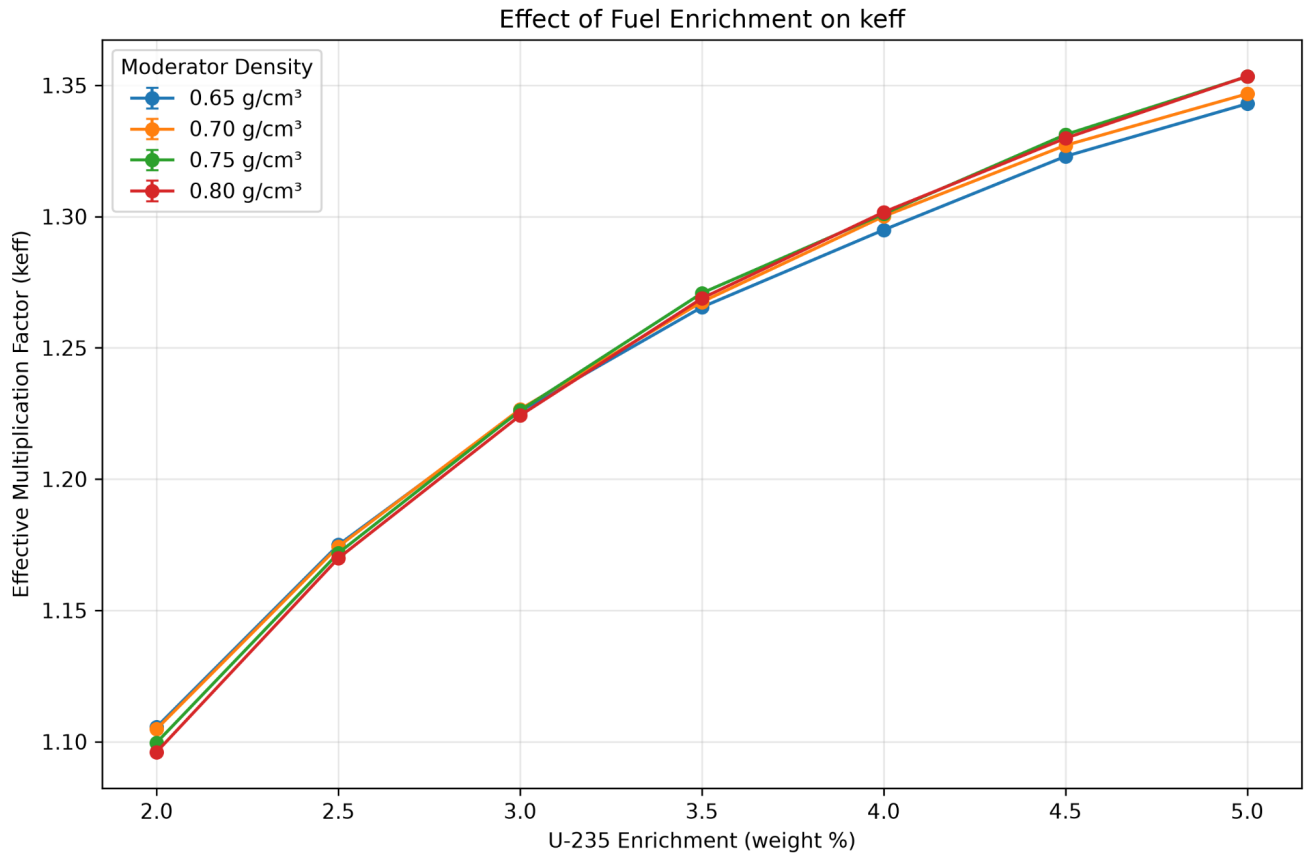


Figure 2. k-effective versus enrichment for all four moderator densities (actual 28-case OpenMC results).

Computed k-effective matrix

U-235 wt%	0.65 g/cm³	0.70 g/cm³	0.75 g/cm³	0.80 g/cm³
2.0	1.10559	1.10483	1.09965	1.09607
2.5	1.17495	1.17432	1.17183	1.16987
3.0	1.22590	1.22662	1.22613	1.22424
3.5	1.26550	1.26748	1.27085	1.26886
4.0	1.29492	1.30008	1.30102	1.30168
4.5	1.32293	1.32712	1.33121	1.32982
5.0	1.34296	1.34675	1.35333	1.35342

The density-response trend was not uniform across enrichment. That observation motivated a separate four-case boron-removal control experiment rather than an unsupported single-cause interpretation. All unrounded values and tally uncertainties are preserved in the CSV archive.

MODERATOR-DENSITY RESPONSE

The density effect was evaluated as Δk = k(0.80 g/cm³) − k(0.65 g/cm³) for each enrichment. It was negative at 2.0% (−0.009515 ± 0.001236) and positive at 5.0% (+0.010456 ± 0.001364). The intermediate response at 3.0% was −0.001659 ± 0.001334, too small relative to sampling uncertainty to establish a precise crossover.

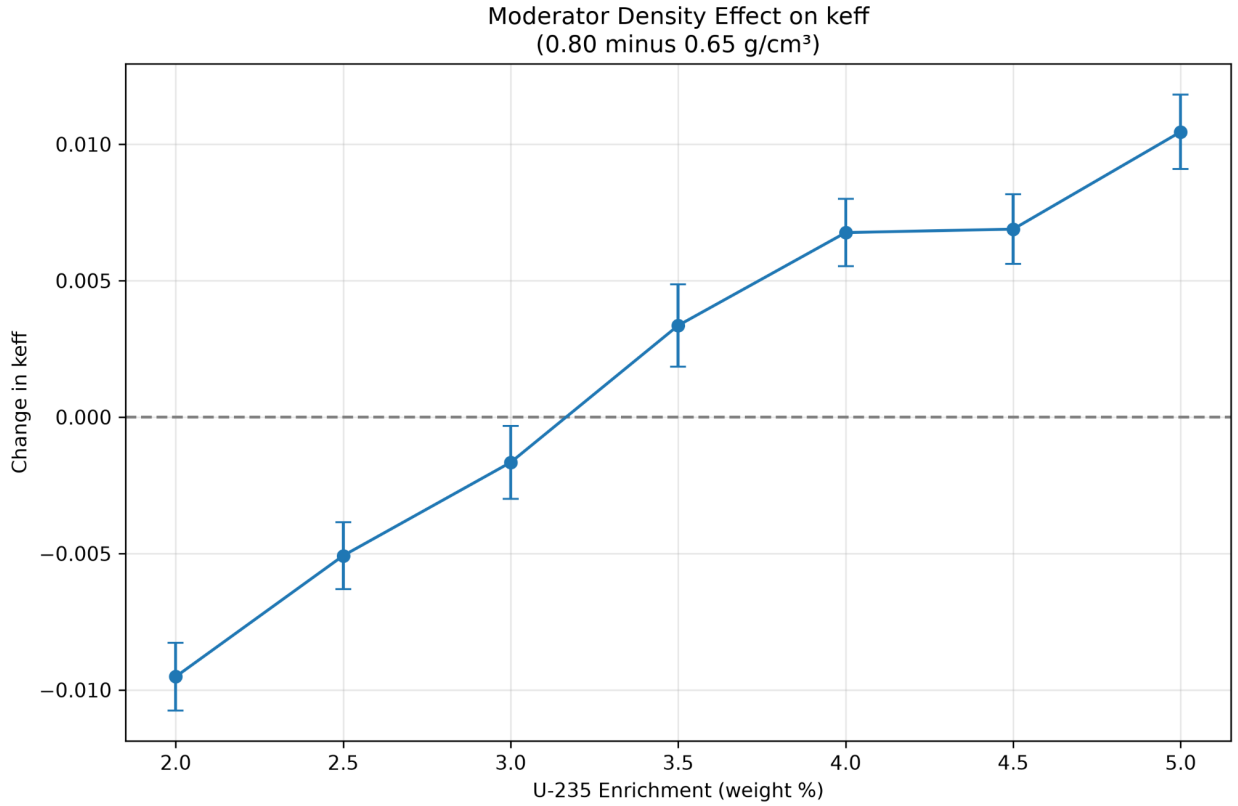


Figure 3. Endpoint moderator-density sensitivity changes sign across the tested enrichment range. The exact transition point is not established.

Boron-removal control

Four additional models removed B-10 and B-11 at 2.0% and 5.0% enrichment, with endpoint moderator densities of 0.65 and 0.80 g/cm³. The total moderator mass density was kept fixed, although normalized hydrogen/oxygen number densities changed slightly when boron was removed.

Enrichment	Borated Δk ± σ	Unborated Δk ± σ
2.0%	−0.009515 ± 0.001236	+0.020118 ± 0.001255
5.0%	+0.010456 ± 0.001364	+0.036469 ± 0.001284

Without boron, the density response was positive at both sampled enrichment endpoints; at 2.0% it reversed sign. This supports a substantial contribution from boron absorption to the original model's density response, while not uniquely isolating every spectrum and material-composition effect.

CONTROL FIGURE, LIMITATIONS AND REPRODUCIBILITY

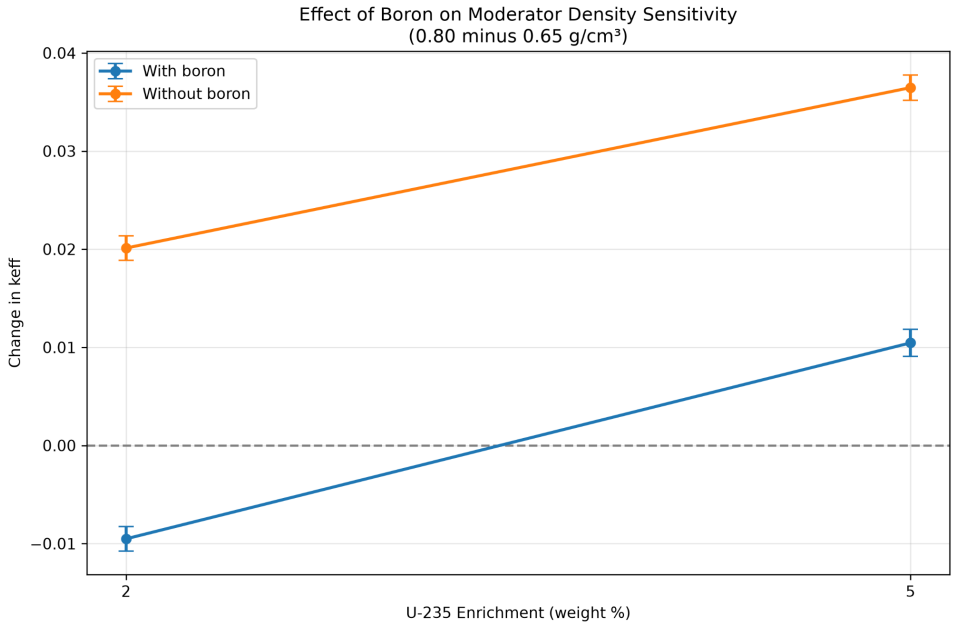


Figure 4. Comparison of the borated and unborated density response at two enrichment endpoints. Error bars propagate independent one-standard-deviation reported uncertainties.

Additional diagnostic

The fuel fission tally increased overall with enrichment (roughly 0.45 to 0.55 reactions per source particle in these runs). The complete figure is included in the repository as figures/fuel_fission_vs_enrichment.png; values are not fissions per second or reactor power.

Scientific limitations

Geometry: reflective infinite pin cell; no full core, explicit fuel gap, leakage, depletion, or thermal hydraulics. Thermal state: density varied independently of temperature using data reported at 294 K; no coupled transient. Composition: boron scaled with moderator density and its removal slightly altered H/O densities. Uncertainty: ± values are sampling errors, not bounds on modeling or nuclear-data error. Data warning: negative Zr-96 probability-table entries were reported; effects were not quantified. Validation: no independent benchmark or experimental comparison.

How to reproduce

The GitHub portfolio contains model/run/analysis scripts, CSV outputs, figures, environment.yml, and README. To regenerate figures, run python analyze_sweep.py and python analyze_boron_control.py. Transport reruns need OpenMC 0.16.0 and ENDF/B-VIII.1 HDF5 data configured via OPENMC_CROSS_SECTIONS. Large HDF5 statepoints are not bundled.

Conclusion

This project provides an end-to-end, reproducible example of nuclear scientific computing: physics-based model construction, automated 28-case simulations, Monte Carlo precision checks, hypothesis-driven control calculations, and transparent communication of limitations. Results describe this simplified model and cannot be extrapolated directly to a real operating reactor.

References

[1] OpenMC PWR pin-cell example: https://github.com/openmc-dev/openmc/blob/develop/openmc/examples.py
[2] OpenMC User Guide - Specifying Tallies: https://docs.openmc.org/en/stable/usersguide/tallies.html
[3] OpenMC User Guide - Simulation Settings (Shannon entropy): https://docs.openmc.org/en/stable/usersguide/settings.html